The data verbs: select · filter · mutate · arrange

Tuesday, June 2, 2026

> **Today: Tuesday June 2**
>
> **Before class**
>
> - **Perusall Assignment**: *DC* §10.1–10.6
>
> **Plan for today**
>
> - Four new data verbs: `filter`, `arrange`, `select`, `mutate`
> - Sorting verbs by *what they touch*: rows vs. columns
> - Combining verbs with the pipe into real pipelines
> - Formatting pipelines so they stay readable
> - Paying off yesterday's IOU: dropping the `NA` penguins
>
> **Helpful links**
>
> - Syllabus · AI policy · Schedule · *R4DS* Ch. 3 · *R4DS* Ch. 4 · *DC* Ch. 10 · dplyr reference

Recap from yesterday

You have the **framework** and your first two data verbs:

Table 1: Where we left off.

Family	Input → Output	The verb that uses it
Reduction	variable → scalar	`summarise()`
Transformation	variable → variable	*…today: `mutate()`*

Family	Input → Output	The verb that uses it
Data verb	data frame → data frame	all of them

`summarise()` is the home of reduction functions. Today you meet `mutate()`, the home of **transformation** functions, plus three verbs that reshape *which* rows and columns you keep.

A map of the verbs

R4DS §3.1.3 organizes dplyr by **what each verb operates on**:

Table 2: The dplyr toolkit.

Touches…	Verbs	What changes
Rows	`filter()`, `arrange()`	which rows, and their order
Columns	`select()`, `mutate()`	which columns, and new ones
Groups	`group_by()`, `summarise()`	*(yesterday)*

> **Tip**
>
> Every one of these obeys the same three rules: **first argument is a data frame, you name columns without quotes, and the output is a brand-new data frame.**

Rows: filter()

filter(): keep the rows you want

`filter()` keeps only the rows that pass a test (*DC* §10.2, R4DS §3.2.1):

```r
penguins |>
  filter(species == "Gentoo")
```

```
# A tibble: 124 x 8
   species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
   <fct>   <fct>           <dbl>         <dbl>             <int>       <int>
 1 Gentoo  Biscoe           46.1          13.2               211        4500
 2 Gentoo  Biscoe           50            16.3               230        5700
 3 Gentoo  Biscoe           48.7          14.1               210        4450
```

```
 4 Gentoo  Biscoe           50          15.2              218       5700
 5 Gentoo  Biscoe           47.6        14.5              215       5400
 6 Gentoo  Biscoe           46.5        13.5              210       4550
 7 Gentoo  Biscoe           45.4        14.6              211       4800
 8 Gentoo  Biscoe           46.7        15.3              219       5200
 9 Gentoo  Biscoe           43.3        13.4              209       4400
10 Gentoo  Biscoe           46.8        15.4              215       5150
# i 114 more rows
# i 2 more variables: sex <fct>, year <int>
```

The test is a **transformation** that returns `TRUE`/`FALSE` for each row; `filter()` keeps the `TRUE`s.

The comparison operators

These are the tests you'll use inside `filter()` (*DC* Table 10.9):

Table 3: Comparisons for `filter()`.

Operator	Means	Example
`==`	equal to	`species == "Gentoo"`
`!=`	not equal to	`sex != "male"`
`> >=`	greater (or equal)	`body_mass_g >= 5000`
`< <=`	less (or equal)	`flipper_length_mm < 190`
`%in%`	is one of	`island %in% c("Dream", "Biscoe")`

Warning

Use `==` to **test** equality, not `=`. A single `=` means "assign," and dplyr will stop you: *"Did you mean `==`?"* This is a classic AI-generated bug too, exactly the kind of thing Friday's lesson said to read for.

Multiple conditions

Separate tests with a comma (or `&`) for **AND**, a vertical bar `|` for **OR** (R4DS §3.2.1):

```
# AND: Adelies that are ALSO on Dream
penguins |>
  filter(species == "Adelie", island == "Dream")
```

```r
# OR: any penguin that is Adelie OR Gentoo
penguins |>
  filter(species == "Adelie" | species == "Gentoo")

# %in% is the clean way to write a long OR
penguins |>
  filter(species %in% c("Adelie", "Gentoo"))
```

> **Note**
>
> Watch the English-trap: `filter(species == "Adelie" | "Gentoo")` does **not** work the way it reads. Each side of `|` must be a complete test.

Paying off yesterday's IOU

Yesterday, `group_by(species, sex)` gave us an `NA` group because some penguins have a missing `sex`. Now we can drop them:

```r
penguins |>
  filter(!is.na(sex)) |>
  group_by(species, sex) |>
  summarise(n = n(), .groups = "drop")
```

```
# A tibble: 6 x 3
  species   sex        n
  <fct>     <fct>  <int>
1 Adelie    female    73
2 Adelie    male      73
3 Chinstrap female    34
4 Chinstrap male      34
5 Gentoo    female    58
6 Gentoo    male      61
```

`is.na(sex)` is `TRUE` for the missing ones; `!` flips it, so `!is.na(sex)` keeps the rows that **do** have a sex. No more `NA` group.

Rows: arrange()

arrange(): set the order

`arrange()` reorders rows without adding or removing any (*DC* §10.4, R4DS §3.2.3). Ascending by default:

```
penguins |>
  arrange(body_mass_g)
```

```
# A tibble: 344 x 8
   species   island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
   <fct>     <fct>              <dbl>         <dbl>             <int>       <int>
 1 Chinstrap Dream               46.9          16.6               192        2700
 2 Adelie    Biscoe              36.5          16.6               181        2850
 3 Adelie    Biscoe              36.4          17.1               184        2850
 4 Adelie    Biscoe              34.5          18.1               187        2900
 5 Adelie    Dream               33.1          16.1               178        2900
 6 Adelie    Torgers~            38.6          17                 188        2900
 7 Chinstrap Dream               43.2          16.6               187        2900
 8 Adelie    Biscoe              37.9          18.6               193        2925
 9 Adelie    Dream               37.5          18.9               179        2975
10 Adelie    Dream               37            16.9               185        3000
# i 334 more rows
# i 2 more variables: sex <fct>, year <int>
```

Wrap a column in `desc()` for descending order, and list several columns to break ties:

```
penguins |>
  arrange(species, desc(body_mass_g))   # by species, then heaviest first
```

Columns: select()

select(): keep the columns you want

`select()` is to columns what `filter()` is to rows (*DC* §10.1, R4DS §3.3.2):

```
penguins |>
  select(species, island, body_mass_g)
```

```
# A tibble: 344 x 3
   species island    body_mass_g
   <fct>   <fct>           <int>
 1 Adelie  Torgersen        3750
 2 Adelie  Torgersen        3800
 3 Adelie  Torgersen        3250
 4 Adelie  Torgersen          NA
 5 Adelie  Torgersen        3450
 6 Adelie  Torgersen        3650
 7 Adelie  Torgersen        3625
 8 Adelie  Torgersen        4675
 9 Adelie  Torgersen        3475
10 Adelie  Torgersen        4250
# i 334 more rows
```

You can grab a range, drop with `!`, or rename as you go:

```r
penguins |> select(species:bill_depth_mm)  # a range of columns
penguins |> select(!year)                   # everything EXCEPT year
penguins |> select(mass = body_mass_g)      # rename while selecting
```

> **Note**
>
> `penguins` has only 8 columns, so `select()` feels optional here. On a 19-column table like `flights` from HW2, or a 76-column one, it's the first thing you reach for.

select() helpers

For wide tables, name columns by pattern instead of one-by-one (R4DS §3.3.2):

```r
penguins |> select(starts_with("bill"))    # bill_length_mm, bill_depth_mm
penguins |> select(ends_with("_mm"))        # all the millimetre measurements
penguins |> select(where(is.numeric))       # every numeric column
```

Columns: mutate()

mutate(): add a new column

`mutate()` adds columns computed from existing ones, keeping every row (*DC* §10.3, R4DS §3.3.1):

```r
penguins |>
  mutate(body_mass_kg = body_mass_g / 1000) |>
  select(species, body_mass_g, body_mass_kg)
```

```
# A tibble: 344 x 3
   species body_mass_g body_mass_kg
   <fct>         <int>        <dbl>
 1 Adelie         3750         3.75
 2 Adelie         3800         3.8
 3 Adelie         3250         3.25
 4 Adelie           NA        NA
 5 Adelie         3450         3.45
 6 Adelie         3650         3.65
 7 Adelie         3625         3.62
 8 Adelie         4675         4.68
 9 Adelie         3475         3.48
10 Adelie         4250         4.25
# i 334 more rows
```

mutate() is where transformations live

This is the parallel *DC* §10.3 draws to yesterday:

> `mutate()` **always** involves a *transformation* function, just as `summarise()` **always** involves a *reduction* function.

```r
# transformation: one value per row → mutate()
penguins |> mutate(bill_ratio = bill_length_mm / bill_depth_mm)

# reduction: one value total → summarise()
penguins |> summarise(avg_bill = mean(bill_length_mm, na.rm = TRUE))
```

> **Tip**
>
> Deciding between them is the same question from yesterday: does the calculation give **one value per row** (`mutate`) or **one value for the whole group** (`summarise`)?

A categorical column with ifelse()

`ifelse()` (from *DC* §7.4) is a transformation that picks between two values per row, perfect inside `mutate()`:

```
penguins |>
  mutate(size = ifelse(body_mass_g > 4000, "large", "small")) |>
  select(species, body_mass_g, size)
```

```
# A tibble: 344 x 3
   species body_mass_g size
   <fct>         <int> <chr>
 1 Adelie         3750 small
 2 Adelie         3800 small
 3 Adelie         3250 small
 4 Adelie           NA <NA>
 5 Adelie         3450 small
 6 Adelie         3650 small
 7 Adelie         3625 small
 8 Adelie         4675 large
 9 Adelie         3475 small
10 Adelie         4250 large
# i 334 more rows
```

Combining verbs

The pipe does the real work

One verb does one thing. Real questions need several, chained with `|>` (R4DS §3.4):

```
penguins |>
  filter(!is.na(sex)) |>
  mutate(body_mass_kg = body_mass_g / 1000) |>
  select(species, sex, body_mass_kg) |>
  arrange(desc(body_mass_kg))
```

Read it top to bottom as English: take penguins, **then** drop missing sex, **then** add mass in kg, **then** keep three columns, **then** sort heaviest first.

> **Tip**
>
> Because each verb returns a new data frame, the output of one is the input to the next. The verbs line up on the left so you can skim the *story* without reading the details.

Putting all six together

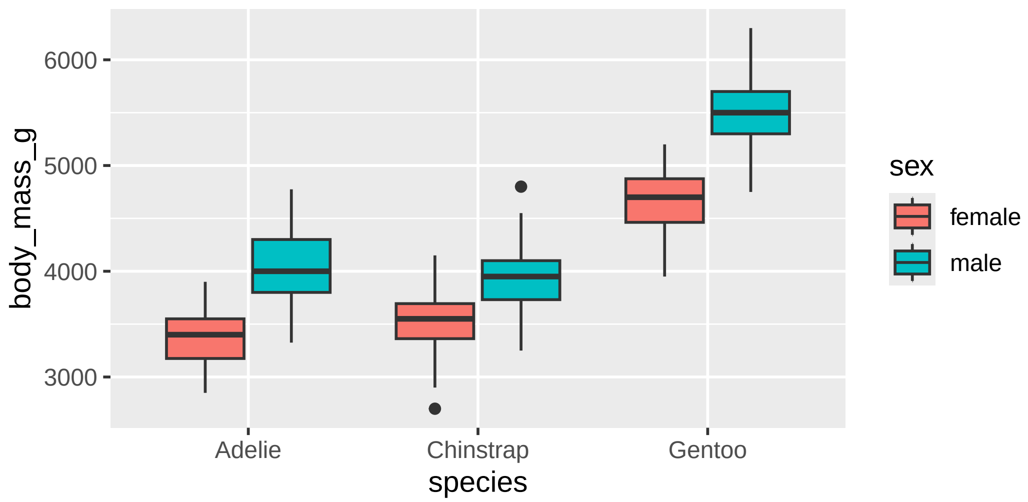
Yesterday's verbs and today's verbs, in one pipeline that answers a real question, "what's the average mass of each species/sex, biggest first?":

```r
penguins |>
  filter(!is.na(sex)) |>
  group_by(species, sex) |>
  summarise(
    n        = n(),
    avg_mass = mean(body_mass_g, na.rm = TRUE),
    .groups  = "drop"
  ) |>
  arrange(desc(avg_mass))
```

```
# A tibble: 6 x 4
  species   sex        n avg_mass
  <fct>     <fct>  <int>    <dbl>
1 Gentoo    male      61    5485.
2 Gentoo    female    58    4680.
3 Adelie    male      73    4043.
4 Chinstrap male      34    3939.
5 Chinstrap female    34    3527.
6 Adelie    female    73    3369.
```

And because the result is just a data frame, pipe it straight into a plot (switching `|>` to `+`):

```r
penguins |>
  filter(!is.na(sex)) |>
  ggplot(aes(x = species, y = body_mass_g, fill = sex)) +
  geom_boxplot()
```



Formatting your pipelines

A multi-step pipeline is only readable if you format it. The rules that matter most (R4DS Ch. 4):

- A space before `|>`, and let `|>` be the **last thing on the line**.
- For verbs with **named arguments** (`mutate`, `summarise`), put each argument on its **own line**, indented two spaces.
- Verbs without named arguments (`filter`, `select`, `arrange`) can stay on one line if they fit.
- Name new columns in `snake_case`: `body_mass_kg`, not `BodyMassKG`.

```
# Strive for
penguins |>
  filter(!is.na(sex)) |>
  group_by(species) |>
  summarise(
    n   = n(),
    avg = mean(body_mass_g, na.rm = TRUE)
  )
```

```
# Avoid
penguins|>filter(!is.na(sex))|>group_by(
species)|>summarize(n=n(),avg=mean(
body_mass_g,na.rm=TRUE))
```

> Tip
>
> Don't hand-format forever: the **`styler`** package reformats a whole file for you. In RStudio, `Cmd/Ctrl + Shift + P` → type "styler". Full guide: R4DS Ch. 4.

A shortcut you'll use constantly

Counting rows per group is so common it has its own verb, `count()` (R4DS §3.2.4):

```
penguins |> count(species)

# is exactly the same as
penguins |>
  group_by(species) |>
  summarise(n = n())
```

> **Note**
>
> `count()` is just `group_by()` + `summarise(n = n())` bundled up. Add `sort = TRUE` to get the biggest groups first.

What we covered today

- **`filter()`** — keep rows that pass a test; `==` not `=`; combine with `,`/`|`/`%in%`; `!is.na()` to drop missing
- **`arrange()`** — reorder rows; `desc()` for descending; extra columns break ties
- **`select()`** — keep/drop/rename columns; `starts_with()` & friends for wide tables
- **`mutate()`** — add columns from transformations; the per-row twin of `summarise()`
- **The pipe** — chain verbs into a readable top-to-bottom story; pipe results into ggplot
- **Style** — space before `|>`, named args on their own lines, `snake_case`, let `styler` do the rest

Activity

Activity: wrangling the penguins

Roughly 25 minutes. Individually or with a neighbor.

- [Activity Canvas Link]

Format: a handout that walks each verb, then asks you to chain them. You'll match tasks to verbs, write `filter`/`arrange`/`select`/`mutate` pipelines on `penguins`, debug a `=` vs `==` error, build one multi-verb pipeline that answers a question, and restyle a deliberately unreadable pipeline.

To turn in: save your `.qmd` and rendered HTML.

Wrap-up & looking ahead

Tomorrow (Wednesday): no new reading, we put all six verbs together on a messier dataset and turn raw rows into summary tables and plots.

Upcoming Deadlines

- **Thu 6/4, before class**: *DC* §12.1
- **Thu 6/4, 11:59 p.m.**: Project: Team formation

Reach out

Stuck? Office hours, or email me (cgc5478@psu.edu) or Xinyue (xpw5228@psu.edu).

Reference

Syllabus · AI policy · Schedule · *R4DS* Ch. 3 · *R4DS* Ch. 4 · *DC* Ch. 10 · dplyr reference